Python String Methods + If-Else

35 Questions, Solutions and Expected Outputs

Topics: isalpha(), isdigit(), isalnum(), islower(), isupper(), isspace(), startswith(), endswith(), len(), and/or/not, and if-else.

How to study: Read the question first, attempt the code, then compare it with the answer and expected output.

Method	What it checks	Example
`isalpha()`	Only alphabetic characters	"Hello".isalpha() → True
`isdigit()`	Only digits	"123".isdigit() → True
`isalnum()`	Letters and/or digits only	"A123".isalnum() → True
`islower()`	All cased letters are lowercase	"hello".islower() → True
`isupper()`	All cased letters are uppercase	"HELLO".isupper() → True
`isspace()`	Only whitespace	" ".isspace() → True
`startswith()`	Beginning of a string	"Python".startswith("Py") → True
`endswith()`	End of a string	"file.py".endswith(".py") → True

Level 1: Basic

Question 1

Check whether "Python" contains only alphabetic characters using `isalpha()`.

```
s = "Python"
if s.isalpha():
    print("Only alphabets")
else:
    print("Not only alphabets")
```

Expected output: Only alphabets

Question 2

Check whether "12345" contains only digits using `isdigit()`.

```
s = "12345"
if s.isdigit():
    print("Only digits")
else:
    print("Not only digits")
```

Expected output: Only digits

Question 3

Check whether "Python123" contains only letters and digits using `isalnum()`.

```
s = "Python123"
if s.isalnum():
    print("Alphanumeric")
else:
    print("Not alphanumeric")
```

Expected output: Alphanumeric

Question 4

Check whether "hello" is lowercase using `islower()`.

```
s = "hello"
if s.islower():
    print("Lowercase")
else:
    print("Not lowercase")
```

Expected output: Lowercase

Question 5

Check whether "HELLO" is uppercase using `isupper()`.

```
s = "HELLO"
if s.isupper():
    print("Uppercase")
else:
    print("Not uppercase")
```

Expected output: Uppercase

Question 6

Check whether " " contains only whitespace using `isspace()`.

```
s = "   "
if s.isspace():
    print("Only whitespace")
else:
    print("Not only whitespace")
```

Expected output: Only whitespace

Question 7

Check whether "Python Programming" starts with "Python".

```
s = "Python Programming"
if s.startswith("Python"):
    print("Starts with Python")
else:
    print("Does not start with Python")
```

Expected output: Starts with Python

Question 8

Check whether "Hello World" ends with "World".

```
s = "Hello World"
if s.endswith("World"):
    print("Ends with World")
else:
    print("Does not end with World")
```

Expected output: Ends with World

Question 9

Check whether "123abc" contains only digits.

```
s = "123abc"
if s.isdigit():
    print("Only digits")
else:
    print("Not only digits")
```

Expected output: Not only digits

Question 10

Check whether "Hello123" is alphanumeric.

```
s = "Hello123"
if s.isalnum():
    print("Alphanumeric")
else:
    print("Not alphanumeric")
```

Expected output: Alphanumeric

Level 2: Intermediate

Question 11

If "python" is lowercase, print Valid; otherwise, print Invalid.

```
s = "python"
if s.islower():
    print("Valid")
else:
    print("Invalid")
```

Expected output: Valid

Question 12

If "WELCOME" is uppercase, print Welcome Message; otherwise, print Invalid Message.

```
s = "WELCOME"
if s.isupper():
    print("Welcome Message")
else:
    print("Invalid Message")
```

Expected output: Welcome Message

Question 13

Check whether "Java123" is alphanumeric. Print Accepted or Rejected.

```
s = "Java123"
if s.isalnum():
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 14

Check whether "student@gmail.com" ends with ".com".

```
s = "student@gmail.com"
if s.endswith(".com"):
    print("Valid Domain")
else:
    print("Invalid Domain")
```

Expected output: Valid Domain

Question 15

Check whether "Python" starts with "Py" and ends with "on".

```
s = "Python"
if s.startswith("Py") and s.endswith("on"):
    print("Both conditions satisfied")
else:
    print("Condition not satisfied")
```

Expected output: Both conditions satisfied

Question 16

Check whether "12345" contains only digits and has exactly five characters.

```
s = "12345"
if s.isdigit() and len(s) == 5:
    print("Valid")
else:
    print("Invalid")
```

Expected output: Valid

Question 17

Check whether "Hello" contains only letters and has a length greater than three.

```
s = "Hello"
if s.isalpha() and len(s) > 3:
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 18

Check whether "HELLO123" is uppercase using `isupper()`.

```
s = "HELLO123"
if s.isupper():
    print("Uppercase")
else:
    print("Not uppercase")
```

Expected output: Uppercase. Digits do not prevent isupper() from returning True when all cased letters are uppercase.

Question 19

Check whether "Python@123" is alphanumeric.

```
s = "Python@123"
if s.isalnum():
    print("Alphanumeric")
else:
    print("Not alphanumeric")
```

Expected output: Not alphanumeric (@ is a special character).

Question 20

Check whether " " contains whitespace and has length greater than zero.

```
s = "   "
if s.isspace() and len(s) > 0:
    print("Valid whitespace")
else:
    print("Invalid")
```

Expected output: Valid whitespace

Level 3: Logical Operators

Question 21

Check whether "Python123" is alphanumeric and has length greater than five.

```
s = "Python123"
if s.isalnum() and len(s) > 5:
    print("Valid")
else:
    print("Invalid")
```

Expected output: Valid

Question 22

Check whether "HELLO" is uppercase and alphabetic.

```
s = "HELLO"
if s.isupper() and s.isalpha():
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 23

Check whether "hello123" is lowercase and alphanumeric.

```
s = "hello123"
if s.islower() and s.isalnum():
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 24

Check whether "12345" contains only digits OR only alphabetic characters.

```
s = "12345"
if s.isdigit() or s.isalpha():
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 25

Check whether "Python.org" starts with "Python" and ends with ".org".

```
s = "Python.org"
if s.startswith("Python") and s.endswith(".org"):
    print("Valid")
else:
    print("Invalid")
```

Expected output: Valid

Question 26

Check whether "Hello World" starts with "Hello" OR ends with "Python".

```
s = "Hello World"
if s.startswith("Hello") or s.endswith("Python"):
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 27

Check whether "123abc" is alphanumeric but not purely numeric.

```
s = "123abc"
if s.isalnum() and not s.isdigit():
    print("Alphanumeric, not purely numeric")
else:
    print("Invalid")
```

Expected output: Alphanumeric, not purely numeric

Question 28

Check whether "WELCOME" is uppercase and exactly seven characters long.

```
s = "WELCOME"
if s.isupper() and len(s) == 7:
    print("Valid")
else:
    print("Invalid")
```

Expected output: Valid

Question 29

Check whether "python" is lowercase but does not start with "j".

```
s = "python"
if s.islower() and not s.startswith("j"):
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Accepted

Question 30

Check whether "12345" contains only digits and is not alphabetic.

```
s = "12345"
if s.isdigit() and not s.isalpha():
    print("Numeric string")
else:
    print("Not a numeric string")
```

Expected output: Numeric string

Level 4: Advanced Conditions

Question 31

A username is valid if it is alphanumeric and has between 5 and 10 characters. Validate "student123".

```
s = "student123"
if s.isalnum() and 5 <= len(s) <= 10:
    print("Valid Username")
else:
    print("Invalid Username")
```

Expected output: Valid Username

Question 32

A filename must start with "report" and end with ".pdf". Validate "report2026.pdf".

```
s = "report2026.pdf"
if s.startswith("report") and s.endswith(".pdf"):
    print("Valid Filename")
else:
    print("Invalid Filename")
```

Expected output: Valid Filename

Question 33

A product code must be alphanumeric, start with "PR", and have exactly eight characters. Validate "PR12AB34".

```
s = "PR12AB34"
if s.isalnum() and s.startswith("PR") and len(s) == 8:
    print("Valid Product Code")
else:
    print("Invalid Product Code")
```

Expected output: Valid Product Code

Question 34

A password for this exercise must be at least eight characters long and alphanumeric. Validate "Python123".

```
s = "Python123"
if len(s) >= 8 and s.isalnum():
    print("Password Accepted")
else:
    print("Password Rejected")
```

Expected output: Password Accepted

Question 35

Accept a string if it contains only alphabetic characters OR only digits. Test "Python", "12345", and "Python123".

```
s = "Python"
if s.isalpha() or s.isdigit():
    print("Accepted")
else:
    print("Rejected")

s = "12345"
if s.isalpha() or s.isdigit():
    print("Accepted")
else:
    print("Rejected")

s = "Python123"
if s.isalpha() or s.isdigit():
    print("Accepted")
else:
    print("Rejected")
```

Expected output: Outputs in order: Accepted, Accepted, Rejected.